SIGN SIGHT: TAMIL AUDIO/VIDEO TO TAMIL SIGN LANGUAGE CONVERSION SYSTEM

(AN AI-POWERED SIGN LANGUAGE CONVERSION SYSTEM FOR ENHANCING ACCESSIBILITY)

Audio/Video – Sign Language Conversion System

Project Final Report

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B.Sc. (Hons) in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

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The above candidate is carrying out research for the undergraduate dissertation under my supervision.

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Signature of co-supervisor Date
(Ms.)

ABSTRACT

This research presents the development of *SignSight*, a machine learning–based system designed to enhance accessibility and communication for the Tamil-speaking deaf and hard-of-hearing community in Sri Lanka and abroad. The focus of this study is the development of a Tamil Audio/Video to Tamil Sign Language (TSL) conversion system, enabling users to input spoken Tamil through audio or video and receive corresponding sign language representations in the form of GIF animations.

The system processes user input by extracting audio from video where necessary, followed by speech-to-text conversion to generate Tamil text. This text is then mapped to predefined Tamil Sign Language gestures stored in a database and displayed through visual animations. The model incorporates Convolutional Neural Networks (CNN) for feature extraction and Long Short-Term Memory (LSTM) networks to improve speech recognition accuracy, particularly in handling pronunciation variations and accents in Tamil speech.

The system was trained using a dataset of over 500 Tamil word-level audio samples. Experimental results indicate a speech-to-text accuracy of 90% and an overall system accuracy of 78%. The system performs well for commonly used words but faces challenges with regional slang variations, unclear audio input, and pronunciation inconsistencies.

This solution contributes toward improving inclusive communication, especially for deaf individuals, cochlear implant users, and the general public interested in learning Tamil Sign Language. Future improvements aim to enhance slang handling, expand datasets, and improve robustness against noise.

Keywords: Tamil Sign Language, Speech-to-Text, CNN, LSTM, Accessibility, Assistive Technology

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LIST OF ABBREVIATIONS

Abbreviation	Description
TSL	Tamil Sign Language
AI	Artificial Intelligence
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
ML	Machine Learning
BSL	British Sign Language
ASL	American Sign Language

1. INTRODUCTION

1.1 Background Study and Literature Review

1.1.1 Background Study

Communication is a fundamental aspect of human interaction, enabling individuals to express ideas, emotions, and information effectively. However, for individuals who are deaf or hard of hearing, communication remains a significant challenge, particularly in environments where sign language is not widely understood. Sign languages serve as a primary medium of communication for such individuals, relying on visual gestures, facial expressions, and body movements instead of spoken language.

Tamil Sign Language (TSL) is used by the Tamil-speaking deaf community in Sri Lanka as well as among Tamil populations worldwide. Despite its importance, TSL has not received sufficient technological support compared to widely recognized sign languages such as American Sign Language (ASL) or British Sign Language (BSL). As a result, communication between hearing individuals and the deaf community often depends on human interpreters or manual learning methods, which can be limited, costly, and inaccessible in many situations.

With the rapid advancement of technology, particularly in the fields of machine learning, speech recognition, and computer vision, there is a growing opportunity to develop intelligent systems that can bridge this communication gap. Speech-to-text technologies and deep learning models have demonstrated significant improvements in recognizing spoken language, even in complex and dynamic environments. These advancements can be leveraged to create assistive systems that convert spoken language into visual sign representations.

This research focuses on the development of a **Tamil Audio/Video to Tamil Sign Language conversion system**, which enables users to input spoken Tamil through audio or video and receive corresponding sign language outputs in the form of GIF animations. By integrating audio processing, deep learning techniques, and visual representation, the system aims to enhance accessibility and inclusivity for the Tamil-speaking deaf community in Sri Lanka and beyond.

1.2 Literature Review

Sign language recognition and translation systems have been widely studied in recent years, particularly with the advancement of machine learning and deep learning technologies. Existing research primarily focuses on three key areas: gesture recognition, speech-to-text conversion, and text-to-sign translation.

Gesture recognition systems commonly utilize computer vision techniques and deep learning models such as Convolutional Neural Networks (CNNs) to identify hand movements and gestures from images or video sequences. These systems have shown promising results in recognizing static and dynamic gestures, particularly in well-resourced sign languages such as ASL. However, many of these systems rely on large datasets and controlled environments, limiting their applicability in real-world scenarios.

Speech recognition systems have also evolved significantly, with models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks being widely used for sequence modeling. LSTM models are particularly effective in capturing temporal dependencies in speech, making them suitable for handling variations in pronunciation and accents. These models have been successfully applied in various languages; however, their performance in low-resource languages like Tamil is still constrained by limited datasets and linguistic diversity.

Text-to-sign translation systems typically map textual input to predefined sign language representations, often using animations or avatar-based systems. While such systems are useful for learning and communication, they often operate at a sentence level and require complex linguistic processing to maintain grammatical correctness. In contrast, word-level mapping approaches offer a simpler and more scalable solution, especially in scenarios where dataset availability is limited.

Despite these advancements, there is a noticeable lack of integrated systems that combine **audio/video input, speech recognition, and sign language output specifically for Tamil Sign Language**. Most existing solutions are either language-specific or lack adaptability to regional variations and slang. Furthermore, many systems rely on external APIs rather than custom-trained models, limiting flexibility and control over performance.

1.3 Research Gap

Based on the analysis of existing literature and current technological solutions, several research gaps have been identified:

- There is a lack of systems specifically designed for **Tamil Sign Language (TSL)**, particularly in the Sri Lankan context.
- Limited availability of **audio/video to sign language conversion systems for Tamil**.
- Existing speech recognition systems struggle with **Tamil pronunciation variations, accents, and regional slang**.
- Insufficient **datasets for Tamil speech and sign language mapping**, leading to reduced model accuracy.
- Most existing systems focus on **gesture recognition or text-based translation**, rather than integrating a complete pipeline from speech to sign output.
- Lack of **simple, scalable word-level mapping systems** that can function effectively in low-resource environments.

These gaps highlight the need for a system that is tailored to Tamil language characteristics while being practical, efficient, and accessible.

1.4 Research Problems

The Tamil-speaking deaf community faces significant barriers in communication due to the absence of automated systems capable of converting spoken Tamil into sign language. Existing solutions either do not support Tamil or fail to address real-world challenges such as pronunciation variability, regional slang, and noisy input environments.

Furthermore, the limited availability of structured datasets and the complexity of Tamil linguistic variations make it difficult to develop accurate and reliable speech recognition systems. As a result, communication between hearing individuals and the deaf community remains inefficient and heavily dependent on human intervention.

Thus, the research problem can be defined as:
"How can a machine learning–based system be designed and implemented to accurately convert Tamil audio and video inputs into Tamil Sign Language representations, while handling challenges such as pronunciation variability, slang, and limited datasets?"

This research problem highlights the need for a solution that is accurate, user-friendly, sustainable, and capable of convenient communication.

1.5 Research Objectives

1.5.1 Main Objective

The primary objective of this research is to design and develop a machine learning–based system capable of converting Tamil audio and video inputs into corresponding Tamil Sign Language (TSL) representations using GIF-based visual outputs. The system aims to enhance communication accessibility for the Tamil-speaking deaf community by providing an efficient and practical translation mechanism from spoken language to visual sign representations.

1.5.2 Specific Objectives

To achieve the main objective, the following specific objectives are defined:

- **To design an efficient audio and video input processing pipeline**
  This objective focuses on developing a system capable of accepting both audio and video inputs from users. For video inputs, the system must extract the audio component and convert it into a format suitable for further processing. The goal is to ensure a unified and consistent input handling mechanism regardless of the input type.
- **To implement audio preprocessing techniques for improved model performance**
  This involves applying preprocessing methods such as noise reduction, normalization, and signal conditioning to enhance the quality of audio input. Since real-world audio may contain background noise and distortions, this step is essential to improve the accuracy of the speech recognition model.

- **To develop a speech-to-text conversion model using CNN and LSTM architectures**
  This objective focuses on building a custom machine learning model that converts Tamil speech into text. The CNN component is used for feature extraction from audio signals, while the LSTM component captures temporal dependencies and sequential

patterns in speech. This hybrid approach aims to improve recognition accuracy, especially for Tamil pronunciation variations.

- **To handle pronunciation variability and accent differences in Tamil speech**
  Tamil language usage varies across regions, with differences in pronunciation, accents, and slang. This objective aims to improve the model's robustness by enabling it to recognize commonly used words despite minor variations in speech patterns.
- To design and implement a word-level text-to-sign mapping mechanism
  This objective involves mapping recognized Tamil text to corresponding Tamil Sign Language representations. A word-level approach is used, where each recognized word is linked to a predefined GIF animation representing the sign. This simplifies implementation while maintaining functional accuracy.
- To develop a GIF-based output system for visual representation of TSL
  This objective focuses on presenting the output in a clear and understandable visual format using GIF animations. The GIFs are derived from sign language videos and custom-created where necessary, ensuring that each word is accurately represented in TSL.
- To evaluate system performance using accuracy metrics and observations
  This includes measuring the performance of both the speech-to-text component and the overall system. Metrics such as recognition accuracy and system reliability are used to assess effectiveness under different conditions, including variations in pronunciation and background noise.
- To identify and analyze challenges related to dataset limitations and linguistic diversity
  This objective involves studying the impact of limited datasets, regional slang, and pronunciation differences on system performance. Understanding these challenges is essential for improving future versions of the system.
- To ensure system scalability and potential for future improvements
  This objective considers the ability to extend the system in the future, including:
    - Expanding the dataset
    - Supporting sentence-level translation
    - Improving noise handling

2. METHODOLOGY

2.1 Overview of the Proposed System

This research focuses on the development of a **Tamil Audio/Video to Tamil Sign Language (TSL) Conversion System** as part of the *SignSight* platform. The system is designed to process user input in the form of audio or video, extract meaningful speech content, and convert it into corresponding Tamil Sign Language representations using GIF-based visual outputs.

The proposed system follows a **modular pipeline architecture**, consisting of the following stages:

1. Input acquisition (audio/video)
2. Audio extraction (for video inputs)
3. Speech-to-text conversion
4. Text processing and mapping
5. GIF-based sign language output generation

Each module is designed to operate independently while maintaining seamless integration within the overall pipeline.

2.2 System Architecture

The system architecture is divided into three main layers:

Input Layer

Accepts:
- Audio files
- Video files

Handles user interaction via frontend (React.js / Next.js)

Processing Layer

- Audio extraction module
- Speech recognition module (CNN + LSTM)
- Text processing and mapping logic

Output Layer

- Displays corresponding **TSL GIF animations** on the frontend

2.3 Input Processing

2.3.1 Audio Input Handling

The system accepts Tamil speech in audio format. The audio input is preprocessed before being passed into the speech recognition model.

2.3.2 Video Input Handling

For video inputs:

- The system extracts audio using preprocessing techniques
- Extracted audio is treated the same as direct audio input

This ensures a unified processing pipeline for both input types.

2.4 Data Preprocessing

Before feeding data into the machine learning model, preprocessing is performed to improve accuracy and consistency.

Steps include:

- Noise reduction (partially implemented – ~50% success)
- Audio normalization
- Segmentation of speech signals
- Conversion into suitable input format for model processing

Noise handling remains a challenge due to variations in recording environments.

2.5 Machine Learning Model Design

The system utilizes a hybrid deep learning approach combining:

2.5.1 Convolutional Neural Networks (CNN)

CNN is used for:

- Extracting spatial features from audio signals
- Identifying patterns in audio representations (such as spectrogram-like features)

2.5.2 Long Short-Term Memory (LSTM)

LSTM is used for:

- Capturing sequential dependencies in speech
- Handling variations in Tamil pronunciation and accents
- Improving contextual understanding of spoken words

The combination of CNN and LSTM enables the model to process both **feature-level and temporal aspects** of Tamil speech effectively.

2.6 Speech-to-Text Conversion

The trained CNN-LSTM model is used to convert processed audio into **Tamil text output**.

Key Characteristics:

- Fully custom-trained model (no external APIs used)
- Handles:
  - Basic Tamil words effectively
  - Pronunciation variations to a certain extent

Limitations:

- Struggles with:
  - Regional slang variations
  - Unclear pronunciation
  - Noisy audio inputs

2.7 Dataset Preparation

The dataset used for training the model consists of:

- **Type:** Tamil audio dataset
- **Content:** Word-level recordings
- **Size:** More than 500 samples

Data Collection Challenges:

- Limited Tamil-speaking population (dataset scarcity)
- Variations in accents and slang
- Inconsistencies in pronunciation across users

These factors directly impacted model generalization.

2.8 Text-to-Sign Mapping

Once Tamil text is generated, it is mapped to corresponding sign language representations.

Key Approach:

- Word-level mapping (not sentence-based)
- Each recognized word corresponds to:

  - A predefined GIF animation

Example:

- Input: *"Amma"* (audio)
- Output: GIF representing *"Amma"* in TSL

Important Note:

- No continuous animation or sentence-level sequencing is implemented
- Each word is processed independently

2.9 GIF Generation and Management

The system uses a collection of GIFs representing Tamil Sign Language gestures.

Sources of GIFs:

- Extracted from existing sign language videos
- Custom-created where required

Storage:

- GIFs are stored and retrieved dynamically during runtime
- Planned integration with MongoDB for efficient storage and retrieval

2.10 System Implementation

I. **Frontend**
   - React.js / Next.js
   - Handles:
     - File upload (audio/video)
     - Display of GIF outputs

II. **Backend**
   - Flask (ML integration)
   - Spring Boot (system-level backend support)

III. **Machine Learning Environment**
   - Python (Jupyter Notebook)
   - Model training and evaluation performed locally

2.11 Testing and Evaluation

The system was evaluated based on:

A. **Accuracy Metrics**
   - Speech-to-text accuracy: **90%**
   - Overall system accuracy: **78%**

B. **Performance Observations**
   - Performs well for:

        - Common words
        - Clearly pronounced speech
    - Performance drops for:
        - Regional slang
        - Unclear audio
        - Complex pronunciation

2.12 Challenges Faced

The development process encountered several challenges:

Dataset Limitations

- Difficulty in collecting diverse Tamil audio samples

Slang Variations

- Words differ regionally (e.g., *"Unavu" vs "Undi"*)
- Model struggles to generalize across variations

Noise Sensitivity

- Background noise affects accuracy
- Current solution partially handles noise (~50%)

2.13 Summary of Methodology

The proposed system successfully integrates audio processing, deep learning, and visual mapping to create a functional Tamil Audio/Video to TSL conversion system. Despite challenges related to dataset limitations and speech variability, the system demonstrates promising results and provides a foundation for further improvements.

3. RESULTS AND DISCUSSION

3.1 Overview

This chapter presents the performance evaluation of the developed Tamil Audio/Video to Tamil Sign Language (TSL) conversion system. The results are analyzed based on accuracy, usability, and observed system behavior under different input conditions

3.2 Results

3.2.1 Speech-to-Text Performance

The speech recognition component, built using a hybrid CNN-LSTM model, achieved an accuracy of:

- **Speech-to-Text Accuracy: 90%**

This indicates that the model is capable of correctly identifying Tamil words from audio input in most controlled scenarios.

3.2.2 Overall System Performance

The complete pipeline—from input processing to GIF output—was evaluated, resulting in:

- **Overall System Accuracy: 78%**

This reduction compared to speech-to-text accuracy is due to compounded errors across stages such as:

- Audio extraction
- Speech recognition
- Word-to-GIF mapping

3.2.3 Functional Performance

The system performs effectively under the following conditions:

- Clearly pronounced Tamil words
- Common and widely used vocabulary
- Minimal background noise

The output GIFs were displayed correctly and were easily interpretable by users.

3.3 Research Findings

From testing and evaluation, the following key findings were identified:

- **Strengths**
  - High accuracy in recognizing **simple and frequently used Tamil words**
  - Efficient **real-time processing pipeline**
  - Successful integration of **audio/video input handling**
  - Effective **visual representation using GIFs**

- **Weaknesses**
  - Difficulty in handling:
    - Regional slang variations
    - Provincial vocabulary differences
  - Performance degradation with:
    - Noisy audio inputs
    - Poor pronunciation
  - Lack of sentence-level interpretation (word-by-word limitation)

3.4 Discussion

3.4.1 Impact of Slang and Language Variability

One of the major challenges identified is the variation in Tamil language usage across regions. For example:

- Standard word: *"Unavu"* (food)
- Regional variation: *"Undi"*

Since the model is trained primarily on standard Tamil words, such variations lead to:

- Misclassification
- Reduced accuracy

This highlights a key limitation in low-resource language processing where datasets lack linguistic diversity.

3.4.2 Effect of Noise on System Performance

Although noise handling techniques were partially implemented, the system currently achieves approximately **50% effectiveness in noise reduction**.

Background noise affects:

- Feature extraction (CNN stage)
- Sequence prediction (LSTM stage)

This results in incorrect or failed speech-to-text conversion.

3.4.3 Word-Level vs Sentence-Level Processing

The system processes inputs at a **word level**, meaning:

- Each word is independently converted into a GIF
- No continuous or contextual animation is generated

While this simplifies implementation and improves accuracy for individual words, it:

- Limits conversational expressiveness
- Reduces natural flow of communication

3.4.4 Dataset Constraints

The dataset consisted of:

- Approx - 500+ Tamil audio samples
- Word-level recordings

Limitations:

- Insufficient diversity in accents
- Limited vocabulary coverage
- Lack of slang representation

These constraints directly impacted the model's generalization capability.

3.5 Summary of Individual Contribution

As part of the *SignSight* system, this research specifically focused on the development of the **Tamil Audio/Video to Tamil Sign Language conversion module**.

Key Contributions:

- Designed and implemented the **audio/video processing pipeline**
- Developed the **speech-to-text conversion system using CNN and LSTM**
- Built the **mapping mechanism from Tamil text to TSL GIF outputs**
- Integrated the full pipeline into a functional system

This component plays a critical role in enabling communication between spoken Tamil users and the deaf community.

4. SUMMARY OF INDIVIDUAL CONTRIBUTION

This research is part of the broader SignSight system, which consists of multiple components developed collaboratively. The focus of this individual contribution is the development of the **Tamil Audio/Video to Tamil Sign Language conversion module**.

The following contributions were made:

- Designed and implemented the **audio and video input processing pipeline**
- Developed the **speech-to-text model using CNN and LSTM architectures**
- Implemented **audio extraction from video inputs**
- Designed the **text-to-GIF mapping mechanism**
- Integrated the full system workflow for end-to-end functionality

This component plays a key role in enabling communication between spoken Tamil users and the deaf community.

5. CONCLUSION

5.1 Conclusion

This research successfully developed a **Tamil Audio/Video to Tamil Sign Language conversion system** as part of the SignSight platform. The system demonstrates the feasibility of using machine learning techniques to bridge the communication gap between hearing and hearing-impaired individuals in the Tamil-speaking community.

The integration of CNN and LSTM models enabled effective speech recognition, achieving a high speech-to-text accuracy of 90%. The overall system achieved 78% accuracy, demonstrating promising results despite challenges related to dataset limitations and linguistic diversity.

The system provides an accessible and scalable solution for:

- Deaf individuals
- Cochlear implant users
- General users learning Tamil Sign Language

By converting spoken Tamil into visual sign representations, the system contributes toward inclusive communication and accessibility.

5.2 Future Work

Several improvements can be made to enhance the system:

1. Dataset Expansion

- Increase dataset size
- Include:
    - Regional slang
    - Different accents
    - Sentence-level data

2. Sentence-Level Translation

- Move from word-based to **sequence-based translation**
- Generate continuous sign animations

3. Improved Noise Handling

- Implement advanced noise reduction techniques
- Use better audio input devices

4. Model Enhancement

- Explore advanced architectures such as:
    - Transformer-based models
    - Attention mechanisms

5. Database Integration

- Integrate MongoDB for efficient storage and retrieval of GIFs

6. Real-Time Deployment

- Optimize system for:
    - Mobile platforms
    - Web-based real-time interaction

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